

EXPLORING ENABLERS AND INHIBITORS OF ELECTRIC VEHICLE ADOPTION INTENTION IN VIETNAM: A DUAL-FACTOR THEORY PERSPECTIVE

Nguyễn Trang Nhung, Khoa Kinh doanh quốc tế - Marketing

Nguyễn Khoa Châu Giang, giangnguyen.31231026719@st.ueh.edu.vn

Trương Huỳnh Như, nhutruong.31231024357@st.ueh.edu.vn

Trần Thanh Phú, phutran.31231022593@st.ueh.edu.vn

Vũ Phương Uyên, uyenvu.31231024398@st.ueh.edu.vn

Abstract

Environmental challenges, particularly climate change and urban air pollution, have elevated the importance of sustainable transportation globally. In Ho Chi Minh City, transportation contributes substantially to particulate matter and greenhouse gas emissions, emphasizing the need for green mobility solutions. Electric vehicles (EVs) offer a promising alternative, yet adoption remains limited in Vietnam due to infrastructural, technological and behavioral constraints. This study addresses the gap in understanding both enablers and inhibitors influencing consumers' EV Adoption Intention. A mixed-methods approach was employed, including a literature review and focus group discussions to refine measurement scales, followed by a structured survey of 460 consumers aged 18 to 45. Data were analyzed using SmartPLS 3, applying reliability assessment, exploratory and confirmatory factor analyses and structural equation modeling. Results reveal that enablers, such as Environmentally Friendly Benefits, Policy Incentives and Driving Enjoyment Benefit, significantly enhance EV Adoption Intention, whereas inhibitors, particularly Infrastructure and Technological Barriers, impede it. Financial Barriers were found less influential due to evolving market conditions and financing options. The findings confirm that enablers and inhibitors operate independently yet simultaneously, consistent with Dual-Factor Theory, underscoring the need to balance positive and negative influences. Furthermore, this research enhances the theoretical knowledge of electric vehicle adoption in emerging markets while offering practical guidance for policymakers and industry players.

Keywords: Electric vehicles, Adoption intention, Enablers, Inhibitors, Dual-Factor Theory

1. Introduction

1.1. Research rationale

Environmental issues, particularly climate change and air pollution, represent critical global challenges, with carbon emissions remaining a major driver of global warming (Alzahrani et al., 2019; J. H. Kim et al., 2020). Following the easing of COVID-19 mobility restrictions, the transportation sector's share of global CO₂ emissions has risen to around 37 percent, largely due to its dependence on fossil fuels, especially in road transport (Pamidimukkala, Patel, et al., 2023; Teter J. Transport tracking report, 2022).

In Vietnam, extensive use of internal combustion engine vehicles has significantly exacerbated urban air pollution, particularly in Ho Chi Minh City, the country's largest economic hub (Chí Nhân, 2025). Cars and motorbikes are identified as the main pollution sources, with PM10 and

PM2.5 levels frequently exceeding national standards in high traffic areas (Giang Anh, 2025). The city's average PM2.5 concentration reaches 31 micrograms per cubic meter, which is six times higher than the World Health Organization guideline, while transportation accounts for about 40 percent of total pollution and emits over 13 million tons of CO₂ annually (Anh Khuê, 2025).

Against this background, Vietnam's commitment to achieving net zero emissions by 2050, announced at COP26, underscores the importance of transitioning toward green transportation (Thùy Dung, 2023). EVs are widely viewed as a promising solution, as previous studies indicate their potential to reduce lifecycle CO₂ emissions by approximately 30 to 50 percent, enhance energy efficiency and alleviate urban air pollution (Hofmann et al., 2016; Y. Liu et al., 2019; Shi et al., 2016). Accordingly, global EV markets have expanded rapidly, particularly in China, Europe and the United States (IEA, 2023; Ruan & Lv, 2023).

However, EV adoption in developing countries, including Vietnam, remains limited due to barriers such as high initial costs, inadequate charging infrastructure, weak policy support and consumer uncertainty toward new technologies (Pamidimukkala, Patel, et al., 2023; Thủy, 2021). In addition, adoption intentions are shaped by psychological and social factors, including resistance to change, trust, awareness, self expression and entrenched consumption habits (Asadi et al., 2022; Murtiningrum et al., 2022; Nguyen, 2021; Yadav & Yadav, 2024).

Overall, investigating EV adoption intentions in Vietnam is both theoretically and practically significant. This study contributes to the limited literature on consumer behavior in emerging markets and offers policy and managerial insights by adopting a dual factor theory perspective that simultaneously examines enabling and inhibiting factors.

1.2. Research objectives

1.2.1. General objective

This study examines both the enablers and inhibitors shaping consumers' intention to adopt EVs in Ho Chi Minh City through the lens of the Dual-Factor Theory, with the aim of generating policy and managerial implications for promoting green transportation.

1.2.2. Specific objectives

- (1) Identify key enablers and inhibitors that influence consumers' intention to adopt EVs in Ho Chi Minh City.
- (2) Assess the significance and relative impact of these factors on EV adoption intention.
- (3) Propose practical and policy recommendations to reduce inhibitors and strengthen enablers of EV adoption.

1.3. Research subject and scope

1.3.1. Research subject

This study examines consumers' intentions to adopt electric vehicles in Ho Chi Minh City, focusing on key enabling and inhibiting factors among individuals aged 18 to 45 with diverse income levels and mobility needs.

1.3.2. Research scope

Spatially, the study is conducted in Ho Chi Minh City.

Temporally, data were collected from July 15, 2025, to September 15, 2025.

1.4. Research questions

- (1) What enablers and inhibitors shape consumers' intention to adopt EVs in Ho Chi Minh City?
- (2) How significant is the relative impact of these enablers and inhibitors on EV adoption intention?
- (3) What practical and policy measures can be proposed to reduce inhibitors and strengthen enablers of EV adoption among consumers in Ho Chi Minh City?

1.5. Research methodology

This study adopts a mixed-methods approach, combining qualitative methods for scale development with quantitative analysis based on 460 valid survey responses collected in Ho Chi Minh City. Data were analyzed using SmartPLS 3, employing reliability tests, factor analysis and structural equation modeling (SEM).

1.6. Structure of the thesis

1. Introduction

This chapter presents the research rationale, objectives, scope, research questions, methodology and overall structure, with a focus on enabling and inhibiting factors influencing EV adoption intention.

2. Theoretical Framework

This chapter reviews relevant theories and prior studies on electric vehicle adoption and introduces the Dual-Factor Theory as the basis for the research model and hypotheses.

3. Research Design

This chapter outlines the research design, including sampling, questionnaire development, data collection, measurement scales and data analysis using SPSS and SmartPLS 3.

4. Research Results

This chapter presents the empirical results, including reliability and validity assessments and structural model analysis, highlighting key enablers and inhibitors of EV adoption intention.

5. Conclusion and Recommendations

This chapter summarizes the main findings, discusses theoretical and practical implications and provides policy and managerial recommendations, along with limitations and future research directions.

2. Theoretical framework

2.1. Conceptual background

EVs are defined as vehicles that operate wholly or partially on electric energy through electric motors and battery storage systems. Depending on their energy sources and operating mechanisms, EVs can be broadly classified into pure electric vehicles, hybrid electric vehicles and fuel-cell electric vehicles (Mwasilu et al., 2014; Richardson, 2013). Owing to their potential to reduce emissions and improve energy efficiency, EVs are increasingly regarded as a central solution for sustainable transportation.

In this context, EV adoption intention is defined as the degree of an individual's willingness to

accept, purchase and use electric vehicles. This concept reflects a psychological state preceding actual behavior and is commonly considered a reliable indicator for predicting future behavior. Drawing on Ajzen's Theory of Planned Behavior, intention plays a mediating role between attitudes, perceptions and actual behavior, while being influenced by various factors related to perceived value, risk perception and policy context. In this study, EV adoption intention is selected as the core dependent variable because actual purchase behavior is often difficult to observe and measure accurately (Ajzen, 1991; Davis et al., 1989; Huijts et al., 2012).

2.2. Theoretical Foundation and Literature Review

The primary theoretical foundation of this study is Dual-factor Theory, originally proposed by Herzberg (1966). According to this approach, the intention to accept or reject a new technology is shaped not only by enabling factors (Enablers) but also simultaneously by inhibiting factors (Inhibitors). These two groups of factors exist independently, may coexist at the same time and are not merely opposite ends of a single continuum (Hsieh, 2016).

Applied to the EV context, enabling factors are typically associated with emotional benefits, functional value, social value and policy support, whereas inhibiting factors mainly stem from technological, financial, environmental and infrastructure barriers. This dual-dimensional perspective allows the study not only to explain why consumers develop intentions to adopt EVs, but also to clarify why they hesitate or postpone their adoption decisions (Cenfetelli, 2004; Cenfetelli & Schwarz, 2010).

Table 2.1. Summary of previous related studies

Author(s)	Year	Context / Scope	Method	Key Factors / Main Findings
Ajzen	1991	General behavioral studies	Conceptual model	Intention is the strongest predictor of actual behavior; attitudes and perceptions shape adoption intention
Huijts et al.	2012	Sustainable energy technologies	Survey, SEM	Attitudes, perceived benefits and social norms significantly affect adoption intention
Rezvani et al.	2015	EV adoption (multiple countries)	Literature review	Financial cost, driving range and charging infrastructure are key barriers; environmental concern acts as a driver
Schuitema et al.	2013	Low-emission vehicles	Survey, regression	Functional value and convenience are strong predictors of EV adoption intention

Higuera-Castillo et al.	2020	Spain	Survey, computational analysis	Driving range, price and monetary incentives are the most influential factors
Giansoldati et al.	2020	Italy	National survey, PCA & cluster analysis	High purchase price, insufficient charging infrastructure and lack of information hinder EV adoption
Illmann & Kluge	2020	Germany	Secondary data analysis	Public charging infrastructure positively affects EV diffusion, but charging speed matters more than quantity
Noel et al.	2020	Nordic countries	Qualitative expert interviews	EV barriers (range, price, charging) are interconnected and strongly linked to consumer knowledge
Wang et al.	2021	Battery EV buyers	Survey, SEM	Environmental concern and social norms outweigh policy perception in shaping purchase intention
Kumar & Alok	2020	Global EV literature	Systematic literature review	Charging infrastructure and cost are well-studied; marketing and dealer experience remain underexplored
Shanmugavel & Micheal	2022	India	Survey at Auto Expo, SEM	Product innovativeness and incentives increase perceived usefulness and purchase intention
Loudiyi et al.	2022	Morocco	Survey, PLS-SEM	Environmental responsibility and policy incentives positively affect perceived value and EV adoption intention
Pamidimukkala et al.	2023	United States	Online survey, SEM	Technological and financial barriers negatively affect EV adoption intention

Source: Compiled by the authors

A review of prior studies indicates that EV adoption intention has been examined from multiple perspectives, including behavioral economics, marketing, public policy and socio-technical systems. Quantitative studies emphasize the roles of cost, driving range, charging infrastructure and financial incentives, while qualitative studies reveal that these barriers are often closely interconnected with consumers' perceptions, experiences and levels of knowledge. However, most existing studies tend to focus on individual groups of factors in isolation, rather than comprehensively examining the simultaneous presence of both enabling and inhibiting forces within a unified analytical framework.

2.3. Conceptual framework and hypotheses development

Based on Dual-factor Theory and the synthesis of previous research, this study proposes an integrated conceptual framework in which EV adoption intention is jointly influenced by two main groups of factors (Herzberg, 1966).

The Enablers include: (i) emotional benefits derived from the driving experience (M.-K. Kim et al., 2018; Schiaroli & Fraccascia, 2025; Venkatesh et al., 2012), (ii) functional value reflected through performance (Bhat et al., 2022; Han et al., 2017; Jain et al., 2021; M. Kang & Park, 2011; W. Zhang et al., 2022; X. Zhang et al., 2013), convenience and product innovativeness (Anderson & Karunamoorthy, 2003; J.-Y. Kang, 2014; Kleijnen et al., 2007; Pura, 2005; Rubio et al., 2013; Sheth et al., 1991; Ustaoglu & Yildiz, 2012), (iii) social value associated with environmentally friendly benefits and environmental concerns (Bickert et al., 2015; Lai et al., 2015; Verma et al., 2018; B. Wang et al., 2017; Wang et al., 2016), (iv) policy incentives (Coffman et al., 2016; Larson et al., 2014; X. Liu et al., 2021; Suxiu et al., 2015; S. Wang et al., 2017) and (v) deal proneness (Atkins & Sae-Young, 2016; Darke & Chung, 2005; Gifford & Comeau, 2011; Lichtenstein et al., 1995). These factors are expected to positively influence EV adoption intention by enhancing perceived value and reducing psychological risk for consumers.

Conversely, the Inhibitors consist of technological (Berkeley et al., 2018; Haddadian et al., 2015; Liao et al., 2017; Patel Ronik et al., 2021; Pelletier et al., 2014), environmental (Gupta & Perveen, 2023; R. Liu et al., 2020; Ramesan et al., 2022), financial (Anastasiadou & Gavanis, 2022; Ghisetti et al., 2017; Hottenrott & Peters, 2009; Lim et al., 2013; Pamidimukkala, Kermanshachi, et al., 2023; Parker et al., 2021; Savignac, 2006) and infrastructural barriers (Illmann & Kluge, 2020; Khan et al., 2023; Li et al., 2020; Ruoso & Ribeiro, 2022). These barriers increase perceptions of inconvenience, cost and uncertainty, thereby weakening consumers' intention to adopt EVs. Importantly, these barriers do not exist independently but may reinforce one another. For example, inadequate charging infrastructure can exacerbate technological concerns and perceived usage costs.

From this theoretical framework, the study develops a set of hypotheses to empirically test the effects of both enabling and inhibiting factors on EV adoption intention among consumers in Ho Chi Minh City. The proposed framework is expected to provide a more comprehensive understanding of behavioral motivations and serve as a basis for policy and managerial implications aimed at accelerating the transition toward sustainable transportation.

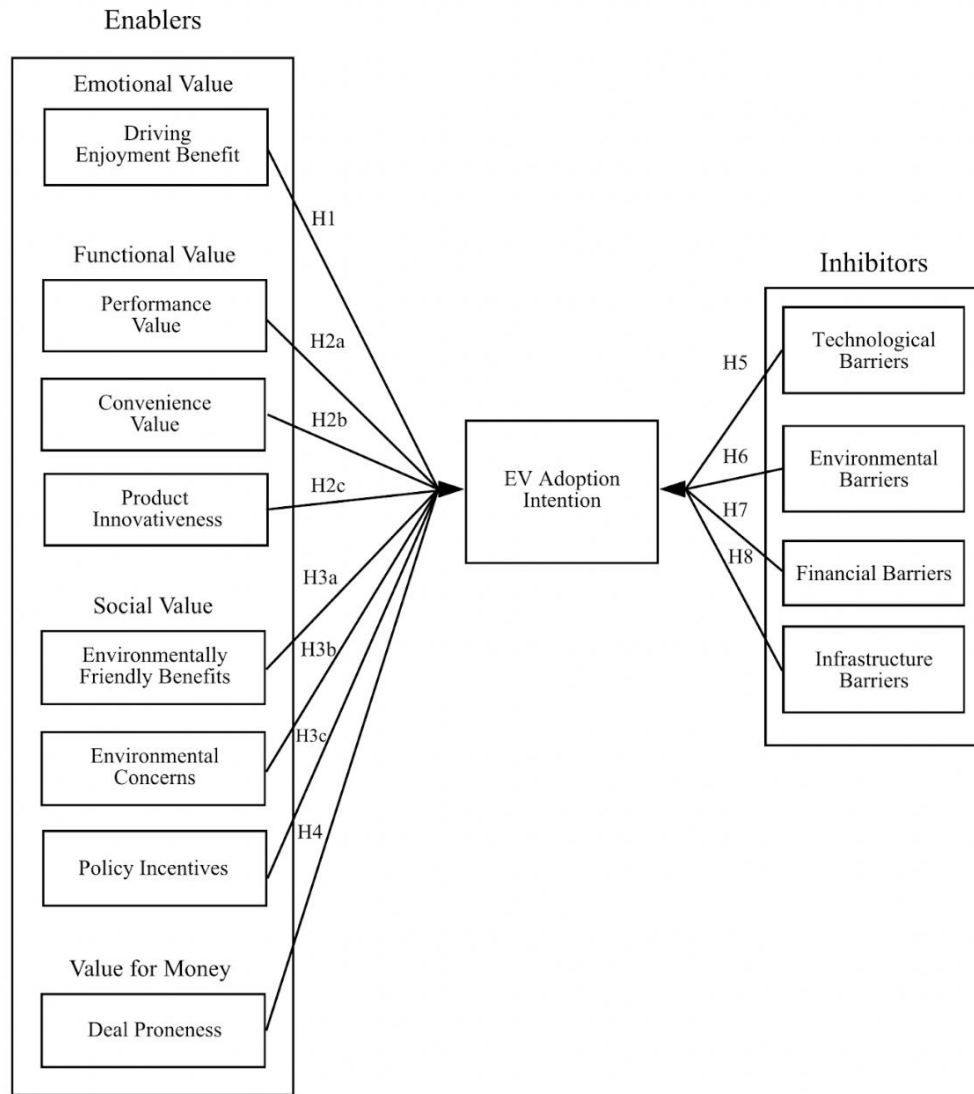


Figure 2.1. Proposed research model

Source: Compiled and developed by the authors

3. Research design

3.1. Research process design

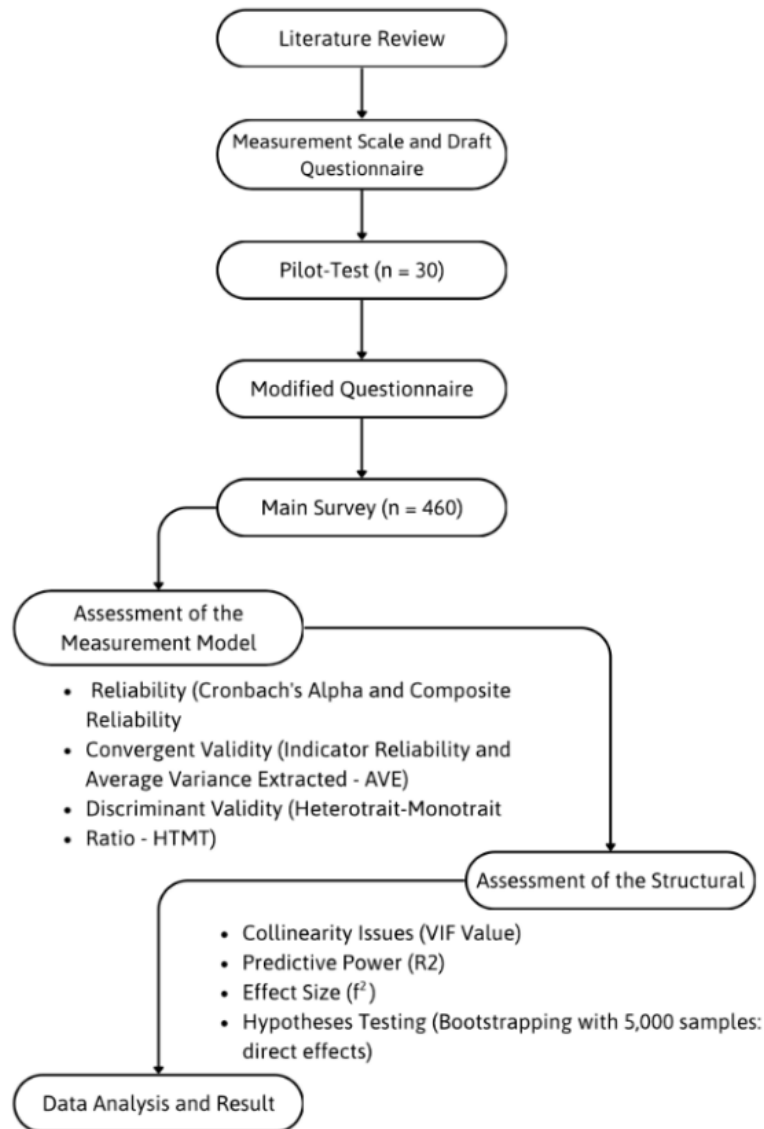


Figure 3.1. Research process

Source: Compiled by the authors

3.2. Quantitative research

3.2.1. Research design

3.2.1.1. Identification of survey subjects

The survey subjects are consumers currently residing in Ho Chi Minh City who possess environmental awareness and consider the possibility of transitioning to electric vehicles. The study focuses on analyzing the perspectives of this specific group.

3.2.1.2. Identification of the survey sample

The survey was conducted with 512 consumers in Ho Chi Minh City and surrounding areas.

3.2.1.3. Determination of sample size

The questionnaire consists of 44 questions; therefore, the minimum sample size was determined

based on the formula $44 \times 10 = 440$ ($n = 440$).

3.2.1.4. Sampling method

This study employs the snowball sampling method, which belongs to the non-probability sampling group. Initial respondents who meet the selection criteria participate in the survey and subsequently refer other individuals with similar characteristics.

3.2.1.5. Sampling approach

The research team collected data indirectly through Google Forms using two approaches: (1) sharing the survey via social media platforms such as Instagram, Zalo and Messenger; and (2) posting the survey in community groups in neighboring districts, clearly stating the research objectives and inviting members to participate.

3.2.1.6. Research model and questionnaire design

Based on previous studies, the research team developed a theoretical model comprising 12 hypotheses and 12 variables and designed a questionnaire containing 44 items. Consistency-check questions were included to enhance data reliability.

3.2.1.7. Analytical tools

The collected data were processed and cleaned using Microsoft Excel, then analyzed using SPSS 27.0 and SmartPLS 3. The results are presented in the form of tables and charts using Microsoft Word.

3.2.2. Research measurement scales

3.2.2.1. Research approach

The questionnaire consists of two sections: (1) demographic information and questions assessing respondents' level of interest in electric vehicles; and (2) questions measuring the variables in the research model. Data were collected using a five-point Likert scale. The survey was primarily distributed to students in areas surrounding Ho Chi Minh City via social media platforms. A total of 460 valid responses were obtained from 512 participants.

3.2.2.2. Measurement scales

Electric Vehicles Adoption Intention			
No.	Encode	Items	References
1	EAI1	Buying an electric car gives me a positive feeling.	Han et al. (2017)
2	EAI2	I plan to choose an electric car when I buy a car in the near future.	
3	EAI3	I am willing to choose an electric car when I buy a car in the near future.	

4	EAI4	I would recommend to others to choose an electric car when they plan to buy a car.	
---	------	--	--

Driving Enjoyment Benefit			
No.	Encode	Items	References
1	ENJ1	EVs give drivers pleasure.	M.-K. Kim et al. (2018)
2	ENJ2	EVs are comfortable with less noise during driving.	
3	ENJ3	EVs are comfortable with low vibration during driving.	

Performance Value			
No.	Encode	Items	References
1	PV1	The electric vehicles possess a consistent quality.	Han et al. (2017)
2	PV2	The electric vehicles in current market are well made.	
3	PV3	The electric vehicles in current market have an acceptable standard of quality	

Convenience Value			
No.	Encode	Items	References
1	CON1	The priority of yearly inspection exclusive for electric vehicles saves my time.	Han et al. (2017)
2	CON2	I think it is convenient since the possession of reserved parking spot.	

Product Innovativeness			
No.	Encode	Items	References
1	PI1	Electric vehicles are highly innovative comparing to the tradition vehicles.	Shanmugavel and Micheal (2022)
2	PI2	Electric vehicles offer unique features for the customers.	

3	PI3	The current technology in electric vehicles are new to the customers.	
---	-----	---	--

Environmentally Friendly Benefit			
No.	Encode	Items	References
1	EFB1	EVs indicate care for the environment.	M.-K. Kim et al. (2018)
2	EFB2	EVs help in responding to global warming via emissions reduction.	
3	EFB3	Driving EVs is a contribution to social responsibility for the environment.	

Policy Incentives			
No.	Encode	Items	References
1	PO1	The government needs to reduce taxes and other bureaucratic costs on electric vehicles.	Loudiyi et al. (2022)
2	PO2	The government must support you financially in the purchase of the electric car.	
3	PO3	The government needs to reduce electricity costs for charging electric.	

Environmental Concern			
No.	Encode	Items	References
1	EC1	I would say of myself that I am environmentally conscious.	Prießner et al. (2018)
2	EC2	Being environmentally friendly is an important part of my personality.	
3	EC3	I would describe myself as someone who cares about the environment.	

Deal Proneness			
No.	Encode	Items	References
1	DP1	Getting a price-off deal when purchasing an EV at	Handarkho et al.

		a physical store makes me feel good.	(2019)
2	DP2	I would purchase EV at a physical store if I could get some discount or rebate from it.	
3	DP3	I like to take advantage of special deals or incentives offered for purchasing EVs at physical stores.	

Technological Barriers			
No.	Encode	Items	References
1	TB1	I believe that electric vehicles still have a limited driving range.	Pamidimukkala, Kermanshachi, et al. (2023)
2	TB2	Long charging time makes using electric vehicles less convenient than traditional vehicles.	
3	TB3	The rapid battery degradation of electric vehicles makes me concerned about their lifespan.	
4	TB4	I think that the number of electric vehicle models available on the market is still limited.	
5	TB5	I am worried about the reliability of electric vehicles.	
6	TB6	The safety issues of electric vehicles remain a concern for me.	

Environmental Barriers			
No.	Encode	Items	References
1	EB1	The disposal of electric vehicle batteries after the end of their lifespan still faces many challenges, which may harm the environment.	Pamidimukkala, Kermanshachi, et al. (2023)
2	EB2	The production process of electric vehicle batteries generates certain negative impacts on the environment.	

Financial Barriers			
No.	Encode	Items	References

1	FB1	I believe that the purchase price of electric vehicles is still too high compared to consumers' affordability.	Pamidimukkala, Kermanshachi, et al. (2023)
2	FB2	I feel that the cost of battery replacement for electric vehicles is a significant financial burden.	
3	FB3	I perceive that the electricity cost for charging electric vehicles remains high, reducing the economic benefit of using them.	
4	FB4	I am concerned that the low resale value of electric vehicles leads to a decline in asset value over time.	
5	FB5	I believe that upgrading or adapting the home electrical system for EV charging is costly.	

Infrastructure Barriers			
No.	Encode	Items	References
1	IB1	I believe that the number of public charging stations is still insufficient to meet the needs of electric vehicle users.	Pamidimukkala, Kermanshachi, et al. (2023)
2	IB2	I find it difficult to charge my electric vehicle when I do not have a garage or a fixed parking space.	
3	IB3	I perceive that maintenance and repair services for electric vehicles are still limited.	
4	IB4	I am concerned that the power grid system for charging electric vehicles is not yet stable and reliable.	

4. Research results

4.1. Sample characteristics

Our study gathered 512 questionnaires from residents in Ho Chi Minh City using various online channels. After a thorough screening, we retained 460 valid responses, roughly 90% of the total, for analysis, discarding invalid ones mainly for incomplete data, inconsistencies, or failing built-in attention checks. Table 4.1 summarizes the sample demographics, showing 58% males and 42% females. The age distribution leaned toward younger adults: 38% were aged 25–31 and 28% fell between 32–38. This aligns well with the main working-age group, which often has steady income and more exposure to new technologies. Most respondents reported monthly incomes in the 10–20 million VND category (38%) or 20–35 million VND (27%), suggesting a largely middle-class urban sample. Overall, the profile of our respondents matches the key demographic for EV

adoption in Vietnam's cities: individuals with financial independence, a keen awareness of environmental issues and an openness to new ideas.

Table 4.1. Demographic Characteristics of Respondents

Variable	Category	Frequency (n)	Percentage (%)
Gender	Male	267	58.0
	Female	193	42.0
Age	18–24 years old	101	22.0
	25–31 years old	175	38.0
	32–38 years old	129	28.0
	39–45 years old	55	12.0
Monthly Income (VND)	Below 5 million	32	7.0
	5 – <10 million	83	18.0
	10 – <20 million	175	38.0
	20 – <35 million	124	27.0
	35 million or above	46	10.0

Total valid respondents: n = 460

Source: Compiled from primary survey data, 2025

4.2. Measurement model assessment

We assessed the measurement model via PLS-SEM, evaluating construct reliability with Cronbach's Alpha, Composite Reliability and Average Variance Extracted. Apart from the Financial Barriers (FB) construct, all others met the established benchmarks: Cronbach's Alpha and CR exceeding 0.70 and AVE above 0.50. However, the Financial Barriers construct showed suboptimal psychometric qualities, including Cronbach's Alpha below 0.70, CR at 0.693, AVE at 0.342 and multiple indicators with loadings under 0.50. In line with Hair et al. (2022), we thus eliminated this construct to enhance the model's overall reliability and validity, while the other inhibitor constructs, Technological Barriers, Environmental Barriers and Infrastructure Barriers, still effectively captured the inhibitory aspects of EV adoption, upholding the Dual-Factor framework's theoretical foundation. Upon re-estimating the model without FB, we confirmed strong internal consistency and convergent validity for the retained constructs.

Table 4.2 Construct Reliability and Validity

Constructs	Cronbach's Alpha	rho_A	CR	AVE
Driving Enjoyment Benefit	0.808	0.816	0.886	0.722
Performance Value	0.799	0.837	0.879	0.708
Convenience Value	0.755	0.765	0.89	0.803
Product Innovativeness	0.822	0.828	0.893	0.736
Environmentally Friendly Benefit	0.805	0.925	0.878	0.708
Environmental Concern	0.803	0.813	0.884	0.717
Policy Incentives	0.818	0.857	0.89	0.729
Deal Proneness	0.812	0.842	0.887	0.724
Technological Barrier	0.909	0.935	0.929	0.685
Environmental Barrier	0.731	0.964	0.871	0.773
Infrastructure Barrier	0.847	0.856	0.897	0.685
Electric Vehicle Adoption Intention	0.845	0.847	0.896	0.682

Source: Compiled from primary survey data, 2025

4.3. Structural model results

Before proceeding to hypothesis testing, our collinearity checks revealed no issues with multicollinearity, as all VIF values remained comfortably under 5.0. For EV Adoption Intention (EAI), the model yielded an R^2 of 0.383 (adjusted $R^2 = 0.368$), meaning it accounts for about 38.3% of the variance in adoption intent, a level of explanatory power that aligns well with norms in consumer behavior studies. To evaluate the significance of these relationships, we performed bootstrapping using 5,000 subsamples, with the outcomes detailed in Table 4.3.

Turning to the enabling factors, the most influential positive drivers on EV adoption intention were Environmentally Friendly Benefits ($\beta = 0.225$), Policy Incentives ($\beta = 0.227$) and Driving Enjoyment Benefits ($\beta = 0.209$), all significant at $p < 0.001$. The remaining enablers—such as Deal Proneness, Product Innovativeness, Performance Value, Environmental Concern and Convenience Value, likewise exhibited meaningful positive associations with EAI. On the flip side, each inhibitor construct registered notable negative effects. Infrastructure Barriers stood out with the

most substantial adverse influence ($\beta = -0.262$, $p < 0.001$), trailed by Technological Barriers ($\beta = -0.170$) and Environmental Barriers ($\beta = -0.163$). Such results underscore how challenges in charging facilities, tech uncertainties and ecological worries continue to pose key hurdles for EV uptake. We did not examine the hypothesis for Financial Barriers (H7), given that this construct was dropped from the final model owing to its insufficient reliability.

Table 4.3. Hypothesis Testing Results (Bootstrapping Analysis)

Hypothesis	Path	β	P Values	Results
H1	DEB $\rightarrow$ EAI	0.209	0.000	Accepted
H2a	PV $\rightarrow$ EAI	0.160	0.000	Accepted
H2b	CV $\rightarrow$ EAI	0.123	0.001	Accepted
H2c	PI $\rightarrow$ EAI	0.163	0.000	Accepted
H3a	EFB $\rightarrow$ EAI	0.225	0.000	Accepted
H3b	EC $\rightarrow$ EAI	0.146	0.000	Accepted
H3c	POLI $\rightarrow$ EAI	0.227	0.000	Accepted
H4	DP $\rightarrow$ EAI	0.183	0.000	Accepted
H5	TB $\rightarrow$ EAI	-0.170	0.000	Accepted
H6	ENB $\rightarrow$ EAI	-0.163	0.000	Accepted
H8	IB $\rightarrow$ EAI	-0.262	0.000	Accepted
H7	FB $\rightarrow$ EAI	-	-	Not tested

4.4. Summary of key findings

Overall, the results provide strong empirical support for the Dual-Factor Theory in the context of EV adoption. Enabling factors related to perceived benefits, policy support and experiential value significantly enhance adoption intention, while infrastructural and technological barriers substantially hinder it. The findings highlight that EV adoption is shaped by the simultaneous interaction of motivators and inhibitors, rather than by positive or negative factors in isolation.

5. Conclusion and recommendations

5.1. Discussion

This study provides comprehensive insights into the factors shaping consumers' intention to adopt EVs in Ho Chi Minh City. Consistent with the Dual-Factor Theory, the findings confirm that adoption intention is jointly influenced by enabling and inhibiting factors. Consumers' decisions are not driven by a single determinant but rather by the balance between perceived benefits and perceived barriers, which is particularly relevant in emerging markets such as Vietnam.

The results indicate that enabling factors exert strong positive effects on EV adoption intention. Among them, Environmentally Friendly Benefits (EFB), Policy Incentives (POLI) and Driving Enjoyment Benefits (DEB) show the strongest influence. This demonstrates that Vietnamese consumers are motivated by both intrinsic values, such as environmental responsibility and emotional satisfaction and extrinsic drivers, including government incentives. Other enablers such as product innovativeness, convenience value, performance value, deal proneness and environmental concern also contribute significantly, highlighting that adoption decisions are formed through a combination of functional, emotional and symbolic evaluations.

Conversely, inhibiting factors have significant negative effects on EV adoption intention. Infrastructure Barriers (IB) emerge as the most critical obstacle, reflecting inadequate charging networks and limited maintenance services. Technological Barriers (TB) further discourage adoption due to concerns regarding battery life, charging time and reliability, while Environmental Barriers (ENB) indicate skepticism about the overall environmental benefits of EVs. Notably, Financial Barriers were excluded from the final model, suggesting that cost-related concerns are becoming less decisive as market conditions and policy support improve.

Overall, the coexistence of strong motivators and barriers reinforces the core proposition of the Dual-Factor Theory: enhancing motivators alone is insufficient if inhibitors remain unresolved. These findings extend the applicability of the Dual-Factor framework to sustainable mobility in emerging economies and provide valuable implications for policymakers and industry stakeholders seeking to accelerate EV adoption.

5.2. Research implications

5.2.1. Theoretical implications

The study strengthens the Dual-Factor Theory by confirming the simultaneous influence of enabling and inhibiting factors on EV adoption intention and extends its application from organizational behavior to green technology acceptance. Furthermore, the findings expand the perceived value theory by demonstrating that consumers evaluate EVs based on functional, emotional and social values rather than purely economic considerations. The dominance of infrastructure barriers over cost or technology barriers highlights the contextual relevance of Vietnam and provides empirical support for the Technology Acceptance Model (TAM) and Theory of Planned Behavior (TPB) in the context of green technology adoption.

5.2.2. Practical implications

From a practical perspective, policymakers should maintain stable incentive policies and prioritize investment in charging infrastructure to build long-term consumer confidence. EV manufacturers should emphasize environmental benefits, enhance driving experience, improve battery technology and strengthen after-sales services. In addition, social and environmental organizations should

collaborate with governments and firms to raise public awareness of sustainable transportation and promote green consumption behaviors.

5.3. Limitations of the study

This study has several limitations. First, the sample was limited to university students in Ho Chi Minh City, which may restrict the generalizability of the findings. Second, the cross-sectional design limits the ability to capture changes in attitudes over time. Third, reliance on self-reported data may introduce response bias. Finally, although the model included multiple motivating and inhibiting factors, other important determinants such as brand image and social influence were not examined.

5.4. Directions for future research

Future studies should expand the demographic and geographic scope to enhance generalizability. Longitudinal or mixed-method approaches are recommended to capture changes in adoption intention over time. Additionally, incorporating new variables and examining post-adoption behaviors would provide deeper insights into the gap between intention and actual EV usage.

5.5. Recommendations

To minimize inhibiting factors, coordinated efforts between government and private sectors are required to improve charging infrastructure, advance battery technology and establish transparent battery recycling systems. To strengthen motivating factors, policymakers and manufacturers should emphasize environmental benefits, ensure policy stability, enhance driving enjoyment, promote product innovation, improve performance transparency, increase convenience and raise environmental awareness among consumers.

REFERENCES

- Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50(2), 179-211. doi:[https://doi.org/10.1016/0749-5978\(91\)90020-T](https://doi.org/10.1016/0749-5978(91)90020-T)
- Alzahrani, K., Hall-Phillips, A., & Zeng, A. (2019). Applying the theory of reasoned action to understanding consumers' intention to adopt hybrid electric vehicles in Saudi Arabia. *Transportation*, 46. doi:10.1007/s11116-017-9801-3
- Anastasiadou, K., & Gavanis, N. (2022). State-of-the-Art Review of the Key Factors Affecting Electric Vehicle Adoption by Consumers. *Energies*, 15, 9409. doi:10.3390/en15249409
- Anderson, R., & Karunamoorthy, S. (2003). E-Satisfaction and E-Loyalty: A Contingency Framework. *Psychology and Marketing*, 20, 123-138. doi:10.1002/mar.10063
- Anh Khuê. (2025). Nồng độ bụi mịn ở TP.Hồ Chí Minh cao vượt 6 lần cảnh báo của WHO. *VnEconomy*. Từ <https://vneconomy.vn/nong-do-bui-min-o-tpho-chi-minh-cao-vuot-6-lan-can-bao-cua-who.htm>
- Asadi, S., Nilashi, M., Iranmanesh, M., Ghobakhloo, M., Samad, S., Alghamdi, A., . . . Mohd, S. (2022). Drivers and barriers of electric vehicle usage in Malaysia: A DEMATEL approach. *Resources, Conservation and Recycling*, 177, 105965. doi:<https://doi.org/10.1016/j.resconrec.2021.105965>
- Atkins, K., & Sae-Young, H. (2016). Smart Shoppers' Purchasing Experiences: Functions of Product Type, Gender and Generation. *International Journal of Marketing Studies*, 8, 1-1. doi:10.5539/ijms.v8n2p1
- Berkeley, N., Jarvis, D., & Jones, A. (2018). Analysing the take up of battery electric vehicles: An investigation of barriers amongst drivers in the UK. *Transportation Research Part D: Transport and Environment*, 63, 466-481. doi:<https://doi.org/10.1016/j.trd.2018.06.016>
- Bhat, F., Verma, M., & Verma, A. (2022). Measuring and Modelling Electric Vehicle Adoption of Indian Consumers. *Transportation in Developing Economies*, 8. doi:10.1007/s40890-021-00143-2
- Bickert, S., Kampker, A., & Greger, D. (2015). Developments of CO2-emissions and costs for small electric and combustion engine vehicles in Germany. *Transportation Research Part D: Transport and Environment*, 36. doi:10.1016/j.trd.2015.02.004
- Cenfetelli, R. (2004). Inhibitors and Enablers as Dual Factor Concepts in Technology Usage. *J. AIS*, 5. doi:10.17705/1jais.00059
- Cenfetelli, R., & Schwarz, A. (2010). Identifying and Testing the Inhibitors of Technology Usage Intentions. *Information Systems Research - ISR*, 21. doi:10.1287/isre.1100.0295
- Coffman, M., Bernstein, P., & Wee, S. (2016). Electric vehicles revisited: a review of factors that affect adoption. *Transport Reviews*, 37, 1-15. doi:10.1080/01441647.2016.1217282
- Chí Nhân. (2025). Vào cao điểm ô nhiễm không khí. *Thanh niên*. Từ <https://thanhnien.vn/vao-cao-diem-o-nhiem-khong-khi-185250109225454773.htm>
- Darke, P.R., & Chung, C.M.Y. (2005). Effects of pricing and promotion on consumer perceptions: it depends on how you frame it. *Journal of Retailing*, 81(1), 35-47. doi:<https://doi.org/10.1016/j.jretai.2005.01.002>
- Davis, F., Bagozzi, R., & Warshaw, P. (1989). User Acceptance of Computer Technology: A Comparison of Two Theoretical Models. *Management Science*, 35, 982-1003. doi:10.1287/mnsc.35.8.982
- Ghisetti, C., Mancinelli, S., Mazzanti, M., & Zoli, M. (2017). Financial barriers and environmental

- innovations: evidence from EU manufacturing firms. *Climate Policy*, 17(sup1), S131-S147. doi:10.1080/14693062.2016.1242057
- Gupta, S., & Perveen, R. (2023). Fuel cell in electric vehicle. *Materials Today: Proceedings*, 79, 434-437. doi:<https://doi.org/10.1016/j.matpr.2023.02.039>
- Giang Anh. (2025). Ôtô, xe máy là nguồn ô nhiễm không khí chính ở TP HCM. *VNExpress*. Từ <https://vnexpress.net/oto-xe-may-la-nguon-o-nhiem-khong-khi-chinh-o-tp-hcm-4918346.html>
- Gifford, R., & Comeau, L. (2011). Message framing influences perceived climate change competence, engagement and behavioral intentions. *Global Environmental Change*, 21, 1301-1307. doi:10.1016/j.gloenvcha.2011.06.004
- Haddadian, G., Khodayar, M., & Shahidehpour, M. (2015). Accelerating the Global Adoption of Electric Vehicles: Barriers and Drivers. *The Electricity Journal*, 28. doi:10.1016/j.tej.2015.11.011
- Han, L., Wang, S., Zhao, D., & Li, J. (2017). The intention to adopt electric vehicles: Driven by functional and non-functional values. *Transportation Research Part A: Policy and Practice*, 103, 185-197. doi:<https://doi.org/10.1016/j.tra.2017.05.033>
- Herzberg, F. (1966). *Work and the Nature of Man*: World Publishing Company.
- Hofmann, J., Guan, D., Chalvatzis, K., & Huo, H. (2016). Assessment of electrical vehicles as a successful driver for reducing CO2 emissions in China. *Applied Energy*, 184, 995-1003. doi:<https://doi.org/10.1016/j.apenergy.2016.06.042>
- Hottenrott, H., & Peters, B. (2009). Innovative Capability and Financing Constraints for Innovation: More Money, More Innovation? *ZEW - Zentrum für Europäische Wirtschaftsforschung / Center for European Economic Research, ZEW Discussion Papers*, 94. doi:10.2139/ssrn.1547083
- Hsieh, P.-J. (2016). An empirical investigation of patients' acceptance and resistance toward the health cloud: The dual factor perspective. *Computers in Human Behavior*, 63, 959-969. doi:<https://doi.org/10.1016/j.chb.2016.06.029>
- Huijts, N.M.A., Molin, E.J.E., & Steg, L. (2012). Psychological factors influencing sustainable energy technology acceptance: A review-based comprehensive framework. *Renewable and Sustainable Energy Reviews*, 16(1), 525-531. doi:<https://doi.org/10.1016/j.rser.2011.08.018>
- IEA. (2023). Executive summary: Electric car sales break new records with momentum expected to continue through 2023. *Global EV Outlook 2023* Từ <https://www.iea.org/reports/global-ev-outlook-2023/executive-summary>
- Illmann, U., & Kluge, J. (2020). Public charging infrastructure and the market diffusion of electric vehicles. *Transportation Research Part D: Transport and Environment*, 86, 102413. doi:10.1016/j.trd.2020.102413
- Jain, N., Bhaskar, K., & Jain, S. (2021). What drives adoption intention of electric vehicles in India? An integrated UTAUT model with environmental concerns, perceived risk and government support. *Research in Transportation Business & Management*, 42, 100730. doi:10.1016/j.rtbm.2021.100730
- Kang, J.-Y. (2014). Augmented reality and motion capture apparel e-shopping values and usage intention. *International Journal of Clothing Science and Technology*, 26, 486-499. doi:10.1108/IJCST-05-2013-0055
- Kang, M., & Park, H. (2011). Impact of experience on government policy toward acceptance of

- hydrogen fuel cell vehicles in Korea. *Energy Policy*, 39, 3465-3475. doi:10.1016/j.enpol.2011.03.045
- Kim, J.H., Kim, N., Moon, H., Lee, S., Jeong, S.Y., Diaz-Pulido, G., . . . Kim, I.N. (2020). Global warming offsets the ecophysiological stress of ocean acidification on temperate crustose coralline algae. *Mar Pollut Bull*, 157, 111324. doi:10.1016/j.marpolbul.2020.111324
- Kim, M.-K., Oh, J., Park, J.-H., & Joo, C. (2018). Perceived value and adoption intention for electric vehicles in Korea: Moderating effects of environmental traits and government supports. *Energy*, 159, 799-809. doi:<https://doi.org/10.1016/j.energy.2018.06.064>
- Kleijnen, M., de ruyter, k., & Wetzels, M. (2007). An assessment of value creation in mobile service delivery and the moderating role of time consciousness. *Journal of Retailing*, 83, 33-46. doi:10.1016/j.jretai.2006.10.004
- Khan, M., Patel, R., Pamidimukkala, A., Kermanshachi, S., Rosenberger, J., Hladik, G., & Foss, A. (2023). Factors that determine a university community's satisfaction levels with public transit services. *Frontiers in Built Environment*, 9. doi:10.3389/fbuil.2023.1125149
- Lai, I., Liu, Y., Sun, X., Zhang, H., & Xu, W. (2015). Factors Influencing the Behavioural Intention towards Full Electric Vehicles: An Empirical Study in Macau. *Sustainability*, 7, 12564-12585. doi:10.3390/su70912564
- Larson, P.D., Viáfara, J., Parsons, R.V., & Elias, A. (2014). Consumer attitudes about electric cars: Pricing analysis and policy implications. *Transportation Research Part A: Policy and Practice*, 69, 299-314. doi:<https://doi.org/10.1016/j.tra.2014.09.002>
- Li, J., Jiao, J., & Tang, Y. (2020). Analysis of the impact of policies intervention on electric vehicles adoption considering information transmission—based on consumer network model. *Energy Policy*, 144, 111560. doi:<https://doi.org/10.1016/j.enpol.2020.111560>
- Liao, F., Molin, E., & van Wee, B. (2017). Consumer preferences for electric vehicles: a literature review. *Transport Reviews*, 37(3), 252-275. doi:10.1080/01441647.2016.1230794
- Lichtenstein, D.R., Netemeyer, R.G., & Burton, S. (1995). Assessing the Domain Specificity of Deal Proneness: A Field Study. *Journal of Consumer Research*, 22(3), 314-326. doi:10.1086/209453
- Lim, M., Mak, H.-Y., & Rong, Y. (2013). Toward Mass Adoption of Electric Vehicles: Impact of the Range and Resale Anxieties. *Manufacturing & Service Operations Management*, 17. doi:10.2139/ssrn.2236560
- Liu, R., Ding, Z., Jiang, X., Sun, J., Jiang, Y., & Qiang, W. (2020). How does experience impact the adoption willingness of battery electric vehicles? The role of psychological factors. *Environmental Science and Pollution Research*, 27. doi:10.1007/s11356-020-08834-w
- Liu, X., Sun, X., Zheng, H., & Huang, D. (2021). Do policy incentives drive electric vehicle adoption? Evidence from China. *Transportation Research Part A: Policy and Practice*, 150, 49-62. doi:<https://doi.org/10.1016/j.tra.2021.05.013>
- Liu, Y., Ouyang, Z., & Cheng, P. (2019). Predicting consumers' adoption of electric vehicles during the city smog crisis: An application of the protective action decision model. *Journal of Environmental Psychology*, 64, 30-38. doi:10.1016/j.jenvp.2019.04.013
- Murtiningrum, A.D., Darmawan, A., & Wong, H. (2022). The adoption of electric motorcycles: A survey of public perception in Indonesia. *Journal of Cleaner Production*, 379, 134737. doi:<https://doi.org/10.1016/j.jclepro.2022.134737>
- Mwasilu, F., Justo, J.J., Kim, E.-K., Do, T.D., & Jung, J.-W. (2014). Electric vehicles and smart grid interaction: A review on vehicle to grid and renewable energy sources integration.

- Renewable and Sustainable Energy Reviews*, 34, 501-516. doi:<https://doi.org/10.1016/j.rser.2014.03.031>
- Nguyen, G.N. (2021). Resistance to Change and Purchase Intention of Electric Vehicles: Empirical evidence from Vietnam. *Asian Journal of Business Research*, 11. doi:10.14707/ajbr.210108
- Pamidimukkala, A., Kermanshachi, S., Rosenberger, J.M., & Hladik, G. (2023). Barriers to electric vehicle adoption: A structural equation modeling analysis. *Transportation Research Procedia*, 73, 305-312. doi:<https://doi.org/10.1016/j.trpro.2023.11.922>
- Pamidimukkala, A., Patel, R., Kermanshachi, S., Rosenberger, J., & Tanvir, S. (2023). *A Review on Shared Mobility and Electric Vehicles*.
- Parker, N., Breetz, H., Salon, D., Conway, M., Williams, J., & Patterson, M. (2021). Who saves money buying electric vehicles? Heterogeneity in total cost of ownership. *Transportation Research Part D: Transport and Environment*, 96, 102893. doi:10.1016/j.trd.2021.102893
- Patel Ronik, K., Etminani-Ghasroodashti, R., Kermanshachi, S., Rosenberger Jay, M., & Weinreich, D. (2021). Exploring Preferences towards Integrating the Autonomous Vehicles with the Current Microtransit Services: A Disability Focus Group Study. In *International Conference on Transportation and Development 2021* (pp. 355-366).
- Pelletier, S., Jabali, O., & Laporte, G. (2014). *Battery electric vehicles for goods distribution: a survey of vehicle technology, market penetration, incentives and practices*.
- Pura, M. (2005). Linking perceived value and loyalty in location-based mobile services. *Managing Service Quality*, 15, 509-538. doi:10.1108/09604520510634005
- Ramesan, S., Kumar, P., & Garg, S. (2022). Analyzing the enablers to overcome the challenges in the adoption of Electric Vehicles in Delhi NCR. *Case Studies on Transport Policy*, 10, 1640-1650. doi:10.1016/j.cstp.2022.06.003
- Richardson, D.B. (2013). Electric vehicles and the electric grid: A review of modeling approaches, Impacts and renewable energy integration. *Renewable and Sustainable Energy Reviews*, 19, 247-254. doi:<https://doi.org/10.1016/j.rser.2012.11.042>
- Ruan, T., & Lv, Q. (2023). Public perception of electric vehicles on Reddit and Twitter: A cross-platform analysis. *Transportation Research Interdisciplinary Perspectives*, 21, 100872. doi:<https://doi.org/10.1016/j.trip.2023.100872>
- Rubio, N., Villaseñor, N., & Yagüe, M.J. (2013). Perceived value of retail service and loyalty to the commercial chain: the role of propensity to buy store brands. *The International Review of Retail, Distribution and Consumer Research*, 23(5), 493-510. doi:10.1080/09593969.2013.832696
- Ruoso, A.C., & Ribeiro, J.L.D. (2022). An assessment of barriers and solutions for the deployment of electric vehicles in the Brazilian market. *Transport Policy*, 127, 218-229. doi:<https://doi.org/10.1016/j.tranpol.2022.09.004>
- Savignac, F. (2006). The impact of financial constraints on innovation : evidence from French manufacturing firms.
- Schiaroli, V., & Fraccascia, L. (2025). Driving the change: How do personal factors and socio-economic context influence electric vehicles adoption across Europe? *Environmental Innovation and Societal Transitions*, 58, 101044. doi:<https://doi.org/10.1016/j.eist.2025.101044>
- Sheth, J., Newman, B., & Gross, B. (1991). *Consumption Values and Market Choices: Theory and Applications* (Vol. 29).

- Shi, X., Wang, X., Yang, J., & Sun, Z. (2016). Electric vehicle transformation in Beijing and the comparative eco-environmental impacts: A case study of electric and gasoline powered taxis. *Journal of Cleaner Production*, 137. doi:10.1016/j.jclepro.2016.07.096
- Suxiu, L., Yingqi, L., & Jingyu, W. (2015, 27-29 July 2015). *Factors affecting the electric vehicle demonstration: 14 international cities/regions cases*. Paper presented at the 2015 International Conference on Logistics, Informatics and Service Sciences (LISS).
- Teter J. Transport tracking report. (2022). Transport. Từ <https://www.iea.org/energy-system/transport>
- Thùy Dung. (2023). Việt Nam có thể đạt được mục tiêu giảm phát thải ròng bằng “0”. *Báo Chính phủ*. Từ <https://baochinhphu.vn/viet-nam-co-the-dat-duoc-muc-tieu-giam-phat-thai-rong-bang-0-102231129162029659.htm>
- Thủy, T. (2021). Tăng chất pin, mở thêm trạm sạc: Mơ chạy ô tô điện xuyên Việt. *VietNamnet*. Từ <https://vietnamnet.vn/tang-chat-pin-mo-them-tram-sac-mo-chay-o-to-dien-xuyen-viet-710186.html>
- Ustaoglu, M., & Yildiz, B. (2012). Innovative Green Technology in Turkey: Electric Vehicles' Future and Forecasting Market Share. *Procedia - Social and Behavioral Sciences*, 41, 139-146. doi:<https://doi.org/10.1016/j.sbspro.2012.04.018>
- Venkatesh, V., Thong, J., & Xu, X. (2012). Consumer Acceptance and Use of Information Technology: Extending the Unified Theory of Acceptance and Use of Technology. *MIS Quarterly*, 36, 157-178. doi:10.2307/41410412
- Verma, V., Chandra, B., & Kumar, S. (2018). Values and ascribed responsibility to predict consumers' attitude and concern towards green hotel visit intention. *Journal of Business Research*, 96, 206-216. doi:10.1016/j.jbusres.2018.11.021
- Wang, B., Shen, Y., & Jin, Y. (2017). Measurement of public awareness of climate change in China: based on a national survey with 4,025 samples. *Chinese Journal of Population Resources and Environment*, 15(4), 285-291. doi:10.1080/10042857.2017.1418276
- Wang, S., Fan, J., Zhao, D., Yang, S., & Fu, Y. (2016). Predicting consumers' intention to adopt hybrid electric vehicles: using an extended version of the theory of planned behavior model. *Transportation*, 43, 1-21. doi:10.1007/s11116-014-9567-9
- Wang, S., Li, J., & Zhao, D. (2017). The impact of policy measures on consumer intention to adopt electric vehicles: Evidence from China. *Transportation Research Part A: Policy and Practice*, 105, 14-26. doi:<https://doi.org/10.1016/j.tra.2017.08.013>
- Yadav, R., & Yadav, R. (2024). Are consumers ready to adopt electric vehicles? Analyzing the barriers and motivators associated with electric vehicle adoption in India: Policy implications for various stakeholders. *Energy Policy*, 190, 114173. doi:<https://doi.org/10.1016/j.enpol.2024.114173>
- Zhang, W., Wang, S., Wan, L., Zhang, Z., & Zhao, D. (2022). Information perspective for understanding consumers' perceptions of electric vehicles and adoption intentions. *Transportation Research Part D: Transport and Environment*, 102, 103157. doi:<https://doi.org/10.1016/j.trd.2021.103157>
- Zhang, X., Wang, K., Hao, Y., Fan, J.-L., & Wei, Y.-M. (2013). The impact of government policy on preference for NEVs: The evidence from China. *Energy Policy*, 61, 382-393. doi:10.1016/j.enpol.2013.06.114

Thông tin chi tiết các tác giả

Họ và tên	Học hàm, học vị	Đơn vị công tác	Địa chỉ	Mobile	Email	Lĩnh vực nghiên cứu
Nguyễn Trang Nhung		Khoa Kinh doanh quốc tế - Marketing	59C Nguyễn Đình Chiểu, phường Xuân Hòa, Thành phố Hồ Chí Minh	0376305567	nhungnguyen.31231026691@st.ueh.edu.vn	Hành vi người tiêu dùng trong bối cảnh chuyển đổi xanh
Nguyễn Khoa Châu Giang		Khoa Kinh tế	59C Nguyễn Đình Chiểu, phường Xuân Hòa, Thành phố Hồ Chí Minh	0822274325	giangnguyen.31231026719@st.ueh.edu.vn	
Trương Huỳnh Như		Khoa Kinh doanh quốc tế - Marketing	59C Nguyễn Đình Chiểu, phường Xuân Hòa, Thành phố Hồ Chí Minh	0866072777	nhutruong.31231024357@st.ueh.edu.vn	
Trần Thanh Phú		Khoa Kinh doanh quốc tế - Marketing	59C Nguyễn Đình Chiểu, phường Xuân Hòa, Thành phố Hồ Chí Minh	0828238767	phutran.31231022593@st.ueh.edu.vn	
Vũ Phương Uyên		Khoa Kinh doanh quốc tế - Marketing	59C Nguyễn Đình Chiểu, phường Xuân Hòa, Thành phố Hồ Chí Minh	0328542994	uyenvu.31231024398@st.ueh.edu.vn	